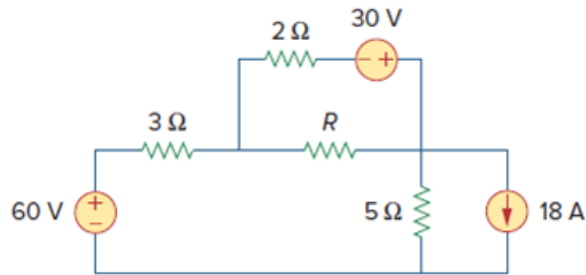


Problem

Find the maximum power that can be delivered to the resistor R in the circuit of Fig. 4.132.

Figure 4.132:



Step-by-step solution

Step 1 of 6

Refer to Figure 4.132 in the text book.

Disconnect the load resistance to calculate the Thevenin's voltage as shown in Figure 1.

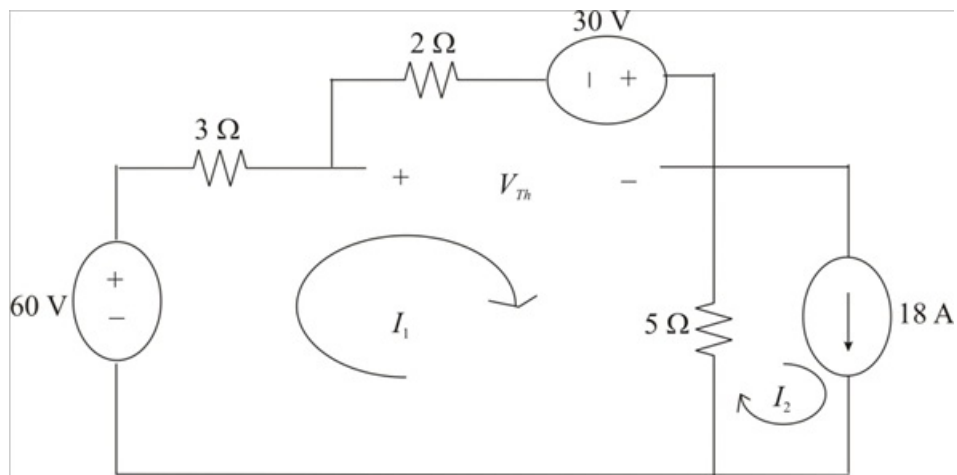


Figure 1

Step 2 of 6

The value of the voltage I_2 is, 18 A .

Apply Kirchhoff's voltage law in loop-1.

$$-60 + I_1(3+2) - 30 + (I_1 - I_2)(5) = 0$$

$$-90 + I_1(3+2+5) - I_2(5) = 0$$

$$-90 + I_1(10) - I_2(5) = 0$$

Substitute 18 A for I_2 .

$$-90 + I_1(10) - (18)(5) = 0$$

$$-90 + I_1(10) - 90 = 0$$

$$I_1(10) - 180 = 0$$

$$I_1(10) = 180$$

$$I_1 = \frac{180}{10}$$

$$= 18\text{ A}$$

Therefore, the value of the current I_1 is 18 A .

Step 3 of 6

Calculate the value of the Thevenin's voltage.

$$V_{\text{Th}} = I_1(2) - 30$$

Substitute 18 A for I_1 .

$$V_{\text{Th}} = (18)(2) - 30$$

$$= 36 - 30$$

$$= 6\text{ V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is 6 V .

Step 4 of 6

Set the values of the sources to calculate the value of the Thevenin's resistance.

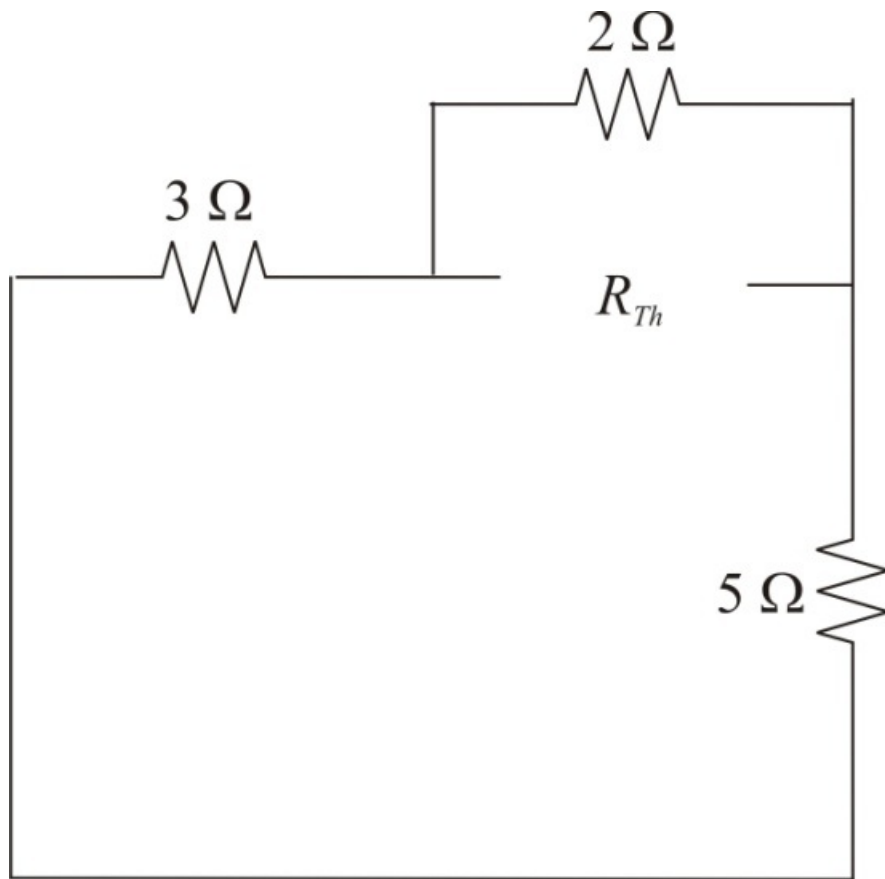


Figure 2

Step 5 of 6

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned} R_{Th} &= 2 \parallel (3+5) \\ &= 2 \parallel 8 \\ &= \frac{(2)(8)}{2+8} \\ &= \frac{16}{10} \end{aligned}$$

$$R_{Th} = 1.6\ \Omega$$

Therefore, the value of the Thevenin's resistance R_{Th} is $1.6\ \Omega$.

For the load to absorb the maximum power, the value of the resistance R must be equal to the Thevenin's resistance $1.6\ \Omega$.

Step 6 of 6

Calculate the maximum power delivered to the load.

$$P_{\max} = \frac{V_{\text{Th}}^2}{4R}$$

Substitute 6 V for V_{Th} and 1.6 Ω for R .

$$\begin{aligned} P_{\max} &= \frac{(6)^2}{4(1.6)} \\ &= \frac{36}{6.4} \\ &= 5.625 \text{ W} \end{aligned}$$

Therefore, the value of the maximum power delivered to the load is 5.625 W.

[Comments \(2\)](#)

☐

Anonymous

Where does the 4 come from in the denominator for Pmax?

☐

Anonymous

its part of the formula to find Pmax
